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Parameter personalisasi : $N=23$, $a=2$, $b=3$, $k=6$, $\theta_0 = 75°$, $d = 0{,}004$

5.1 Limit Aljabar

a. $L_1 = \lim (x \to k) \dfrac{(x^2-k^2)}{(x-k)}$

$L_2 = \lim (x \to 0) \dfrac{(\sqrt{(x+(k+1)^2)} - (k+1))}{x}$

$L_3 = \lim (x \to \infty) \dfrac{((a+2)x^3 + b \cdot x + 7)}{(x^3 - x + k)}$

Jawab :

$k=6$ $L_1 = \lim x \to 6 \dfrac{x^2-36}{x-6} = \dfrac{(x-6)(x+6)}{x-6} = x+6$

$x = 6 \to x+6 = 6+6 = 12$; $L_1 = 12$ //

$L_2 = \lim x \to 0 \dfrac{\sqrt{x+(6+1)^2} - (6+1)}{x}$

$= \lim x \to 0 \dfrac{\sqrt{x+(7)^2} - 7}{x} = \dfrac{\sqrt{x+49} - 7}{x} \times \dfrac{\sqrt{x+49}+7}{\sqrt{x+49}+7}$

$= \dfrac{1}{\sqrt{x+49}+7}$ subs : $\dfrac{1}{7+7} = \dfrac{1}{14}$; $L_2 = \dfrac{1}{14}$ //

$L_3 = \lim x \to \infty \dfrac{(2+2)x^3 + 3x + 7)}{(x^3 - x + 6)} = \dfrac{4x^3+3x+7}{x^3-x+6}$ dibagi x^3

$= \dfrac{4+0+0}{1-0+0} = \dfrac{4}{1} = 4$; $L_3 = 4$ //

5.2 Limit Trigonometri & Model Seasonal

c. $L_4 = \lim x \to 0 \dfrac{\sin((a+1) \cdot x) \cdot \tan((b+2) \cdot x)}{x^2}$

$a = 2$,

$b = 3$

$= \lim x \to 0 \dfrac{\sin(3x) \tan(5x)}{x^2}$

$= \lim x \to 0 \left(\dfrac{\sin(3x)}{x}\right)\left(\dfrac{\tan(5x)}{x}\right) = 3 \times 5 = 15$; $L_4 = 15$ //

Pendekatan ini sah karena menggunakan limit dasar trigonometri : $\lim x \to 0 \dfrac{\sin x}{x} = 1$ dan $\lim x \to 0 \dfrac{\tan x}{x} = 1$. Sehingga hasil limit dapat dihitung.

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5.3 Limit tak hingga & konvergensi CLV

d). $R(n) = \dfrac{100 \cdot n + k}{n + (a+b+1)}$ n = bulan ke n $\lim n \to \infty$

$$R(n) = \frac{100n + 6}{n + (2+3+1)}$$

$$= \frac{100n + 6}{n + 6}$$

$$= \frac{100 + 6/n}{1 + 6/n}$$

limit = 100 // Pendapatan stabil yaitu 100 ribu/bulan.

e). $\lim n \to \infty \left(\sqrt{n^2+6n} - \sqrt{n^2+3n}\right) \times \dfrac{\sqrt{n^2+6n} + \sqrt{n^2+3n}}{\sqrt{n^2+6n} + \sqrt{n^2+3n}}$

$$= \frac{3n}{\sqrt{n^2+6n} + \sqrt{n^2+3n}}$$

$$= \frac{3}{\sqrt{1+6/n} + \sqrt{1+3/n}}$$

Limit $= \dfrac{3}{1+1} = \dfrac{3}{2} = 1{,}5$ // Perbedaan 2 model, mendekati konstan (1,5).

5.4 Kekontinuan fungsi Biaya Piecewise

f). fungsi :

$$C(x) = \begin{cases} \dfrac{x^2 - k^2}{x - k}, & x < k \\ p, & x = k \\ ax + q, & x > k \end{cases}$$

Syarat kontinuitas :

fungsi kontinu di $x = k$ jika $\lim x \to k^- \; C(x) = C(k) = \lim x \to k^+ \; C(x)$

artinya :

1. limit kiri ada
2. limit kanan ada
3. Nilai fungsi sama

limit kiri :

$\lim x \to k^- \dfrac{x^2 - k^2}{x - k}$

$x^2 - k^2 = (x-k)(x+k)$

$= \dfrac{(x-k)(x+k)}{x-k} = x + k$

$x = k = 6$

$6 + 6 = 12$

$\lim x \to 6^- \; C(x) = 12$ //

$C(6) = p$

$p = 12$

limit kanan :

$\lim x \to 6^+ (ax + q) = a(6) + q = 6a + q$

$a = 2 = 12 + q$

limit kiri = limit kanan

$12 = 12 + q$

$\boxed{q = 0 \quad p = 12}$

$\lim x \to 6^- = 12$

$C(6) = 12$

$\lim x \to 6^+ = 12$

$$C(x) = \begin{cases} \dfrac{x^2 - 36}{x - 6}, & x < 6 \\ 12, & x = 6 \\ 2x, & x > 6 \end{cases}$$

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